



— TECHNICAL DATA SHEET

NN-RockBond Cement Capsule

Cement Capsule — Grout Material for Roof Bolting in Mines



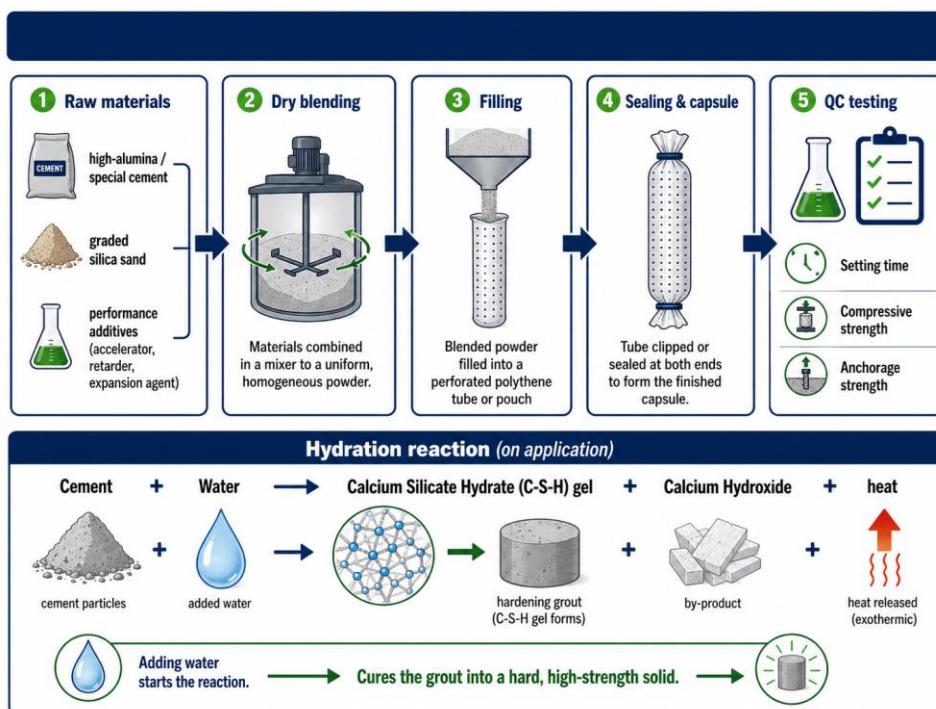
Pre-packed cement grout capsules, supplied in moisture-protected cartons

Document No.	Revision	Issue Date	Product Code
NNPL/TDS/PROD-05	1.0	27 June 2026	CC-05

01 PRODUCT DESCRIPTION

NN-RockBond Cement Capsule is a pre-packed cementitious grout capsule used as a grouting medium for roof bolting and strata control in underground coal mines and tunnels. Each capsule contains a precisely proportioned blend of high-alumina / special cement, graded sand and performance additives, sealed in a perforated polythene pouch.

In use, the capsule is soaked in water for a short period and inserted into the drilled borehole; the roof bolt is then spun through the capsule, rupturing the pouch and mixing the grout to form a rapid-setting, high-strength anchor. The product is designed to comply with the testing and performance requirements of DGMS Circular (Approval) No. 05 of 2013 for cement capsules used as grout material for roof bolting in mines.



Manufacturing flow and on-application hydration reaction (cement + water → C-S-H gel)

02 KEY FEATURES & BENEFITS

- **DGMS-compliant** — Formulated to meet DGMS Circular (Approval) No. 05 of 2013 for cement-capsule grout in below-ground coal mines.
- **Rapid setting** — Fast soaking and setting allow quick installation and early load transfer in the bolting cycle.
- **High anchorage** — Develops high anchorage and compressive strength for reliable roof support.
- **Non-flammable & inorganic** — Cementitious grout is non-flammable — a safety advantage over resin systems in fiery mines.
- **No on-site mixing** — Pre-packed and pre-proportioned; only water soaking is required, eliminating site batching errors.
- **Corrosion-protective** — The set cement grout encapsulates and protects the steel bolt from corrosive mine water.

03 PHYSICO-MECHANICAL PROPERTIES

Per DGMS Circular (Approval) No. 05 of 2013, Para 2.0 (A).

Parameter	Standard
Soaking time of the cement capsule in water	Maximum 3 minutes
Initial setting time (including soaking time)	Minimum 5 minutes – Maximum 10 minutes
Final setting time (including soaking & initial setting)	Maximum 15 minutes
Compressive strength gain — after 30 minutes	≥ 3.0 MPa
Compressive strength gain — after 2 hours	≥ 7.0 MPa
Compressive strength gain — after 24 hours	≥ 12.0 MPa
Compressive strength gain — after 7 / 21 / 28 days	≥ 12.0 MPa
Anchorage strength gain — after 30 minutes	3.0 tonnes



Parameter	Standard
Anchorage strength gain — after 2 hours	5.0 tonnes
Anchorage strength gain — after 24 hours	10.0 tonnes
Anchorage strength gain — after 28 days	12.0 tonnes

04 CHEMICAL PROPERTIES

Per DGMS Circular (Approval) No. 05 of 2013, Para 2.0 (B).

Parameter	Standard
Chloride	Maximum 0.1 %
Shrinkage	No shrinkage
Expansion (Autoclave method)	Maximum 0.8 %
Expansion (Le Chatelier method)	Maximum 10 mm
Sulphuric anhydrite	Maximum 15 %

05 APPLICATION & INSTALLATION



Roof-bolt installation into the strata in an underground coal mine

- **Drill** — Drill the borehole to the specified depth and diameter; ensure it is clean and free from obstruction.
- **Soak** — Soak the required number of capsules in clean water (maximum 3 minutes, until air release / bubbling stops).
- **Insert** — Insert the soaked capsule(s) into the borehole to achieve full-column grouting.
- **Spin & anchor** — Spin the roof bolt through the capsule to rupture the pouch, mix the grout and seat the bolt.
- **Cure** — Allow to set and develop strength; do not disturb the bolt until the required anchorage is achieved.



06 PACKAGING, STORAGE & SHELF LIFE

Item	Detail
Capsule sizes	Available in standard borehole diameters and lengths to suit the bolt / hole (made to order)
Packaging	Capsules packed in perforated polythene pouches; supplied in moisture-protected cartons
Storage	Store in a cool, dry, covered area protected from moisture, water and direct weather; rotate stock (FIFO)
Shelf life	6 months from date of manufacture in unopened, moisture-protected packaging
Handling	Protect from extreme weather and moisture, which can prematurely activate or degrade the grout

07 COMPLIANCE & TESTING

Each batch is manufactured and tested in line with DGMS Circular (Approval) No. 05 of 2013. Testing of soaking / setting time, compressive strength (at 30 min, 2 h, 24 h, 7, 21 and 28 days), anchorage strength and chemical parameters is carried out per the methods in Annexure-1 of the circular (referencing IS 4031 and IS 4032). Batch records, usage and storage manuals are maintained and supplied with consignments as required by the circular.

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